

Chapter 10

AI Literacy from Educators' Perspectives



After understanding AI literacy from the perspective of human-design factor, this chapter presents a conceptual framework introducing an expanded view of AI literacy from educators' perspectives. It moves beyond technological competencies and tries to identify a more holistic and broader understanding about AI literacy education. When using these novel AI tools to teach, educators need to be equipped with adequate technological literacy skills and knowledge. In this way, they can teach AI literacy and promote other digital competencies such as collaboration and communication among their students in AI-driven environments. Since teachers may not have rich technical knowledge to apply AI educational applications to facilitate their teaching. As one of the most important twenty-first-century competencies, AI literacy can be conceptualized as the knowledge, skills, and attitudes necessary to be competitive in the twenty-first-century workforce. Teacher education and professional development should be reworked to incorporate training in teaching key digital competencies.

10.1 Understanding Teachers' AI Digital Competencies

Artificial intelligence technology has gained its popularity in education in recent years. It reshapes education which helps educators to automate nonteaching tasks to reduce their workload and enhance learning analysis and teaching optimization (Kexin et al., 2020). Further, it helps promote students' personalized learning (Ahmad et al., 2022) and advance students' knowledge acquisition using AI-empowered systems (e.g., Chen et al., 2020; Hwang et al., 2020). However, these technologies are novel to educators who may not have enough knowledge and skills to manipulate these AI-driven educational applications for learning and teaching purposes (Ally, 2019; Seo et al., 2021). They may not have rich technological experience to design AI-driven learning environments, construct learning analysis,

and set the rules in various AI-enhanced systems to automate responses, assignments, and feedback (Seo et al., 2021), not to mention developing students' AI literacy (Yau et al., 2022). Moreover, challenges were identified such as AI-based misunderstandings and misleadingness, limitations, and hidden ethical issues behind different platforms (Akgun & Greenhow, 2021; Sijing & Lan, 2018).

To develop student' AI literacy, educators need to upskill and reskill their knowledge to design relevant instruction (e.g., pedagogy, content, activities) for their students (Williamson & Eynon, 2020). Research examples were used to identify what types of AI digital competencies should be emphasized for teachers. A set of teachers' AI digital competencies is proposed based on the P21's Framework for the 21st Century Learning. This chapter aims to extend the proposal of viewing AI literacy as an important twenty-first-century skill set (Ng et al., 2021a, b). As mentioned in Chap. 4, AI literacy moves toward a broader understanding that identifies other nontechnical, critical, and complex literacy that K–16 learners need to learn to manipulate AI technologies ethically and effectively. Toward a holistic picture, teachers should not view AI literacy as an independent domain but an avenue to develop other important skill sets such as life and career skills, multidisciplinary skills, learning, and innovation skills. With the P21 framework, teachers can use it as a guideline to equip themselves with necessary digital competencies to facilitate better instruction (Lindfors et al., 2021).

Several digital competency frameworks have been used in teacher education, such as DigCompEdu (Caena & Redecker, 2019) and the International Society for Technology in Education (ISTE) standards (Crompton, 2017) to update the necessary digital competencies for today's digital world. These frameworks inform educators how AI should be incorporated in their subject areas and levels of study. According to the DigCompEdu standards, teachers' digital competencies can be categorized in six areas that facilitate their professional activities (Misthou & Paliouras, 2022): (1) professional engagement, using AI for communication, collaboration, and professional development; (2) resource management, sourcing, creating, and sharing AI-empowered resources; (3) teaching and learning, managing the use of AI for teaching/learning; (4) assessment, using AI to enhance automatic assessment and analysis; (5) empowering learners, Using AI to enhance inclusion, personalization, and learning engagement; and (6) facilitating learners' AI competence, enabling learners to create their own intelligent agents responsibly and use AI for information sharing, communication, content creation, and problem-solving. The ISTE standards consider AI as a new digital competence requirement to develop citizens' necessary knowledge, skills, and attitudes (Riina et al., 2022). These frameworks offer a common understanding across the European Union and beyond of what digital competence needs to frame digital skills policy, curricula development, and assessment of digital skills (Riina et al., 2022).

Among these models, this Chapter proposes the P21's Framework for the 21st Century Learning illustrates the knowledge and skills that educators need to succeed in working, learning, and living (National Research Council, 2012). The framework is famous for educators and business leaders to identify learning standards for implementing curriculum, instruction, assessments, and learning

environments. This chapter does not focus on the technical perspectives for fostering students' AI literacy. Instead, it aims to focus on broader digital competencies (e.g., communication, collaboration, critical thinking, life, and career skills) that support teachers to conduct teaching using AI technologies and implement AI literacy education. In this way, teachers should not only know and use AI applications to empower their students and prepare related teaching resources. They should also master other AI digital competencies such as applying AI knowledge in their disciplines and life and career skills to prepare students to enter the future workforce and communicating and collaborating with their peers.

10.2 Essential AI Digital Competencies for Educators (P21)

Recent scholars proposed the need to reconsider teachers' professional development programs to meet the educational standards and expectations in the new digital era. For example, Trujillo-Cabezas (2020) suggested some teacher's AI digital competencies such as using basic applications, managing information, creating learning content, and connecting their students via technology. Ng et al. (2021b) incorporated AI literacy development into the TPACK model which guides teachers to consider related teaching content, pedagogy, and technology. Using the P21's Framework, this section investigates what teacher competencies are desired for effective online teaching with AI technology. Four key competencies are identified. The key competencies of teachers should focus not only on the acquisition of basic AI knowledge and skills but also on the cultivation of the capabilities necessary for the teacher's adaptation to, survival in, and control over the future society and their lifelong professional development.

Core Subjects, 3Rs, and Twenty-First-Century Themes To be a successful student and adult in the future, educators have a role to help students to master the core subjects and twenty-first-century themes. AI has commonly become a topic in computer science education, but it has already become part of these core subjects of education. In schools, teachers are from diverse educational backgrounds and subjects, and they need to collaborate and work together to adopt, adapt, and implement interdisciplinary materials and approaches. Artificial intelligence technologies could be new to most teachers, and it takes effort for educators to incorporate AI into their knowledge domains and other twenty-first-century themes such as global awareness and financial, environmental, civic, and health literacy. For example, Ng et al. (2022) designed a digital story writing program to develop students' AI concepts, and students can use these concepts to write their stories. When writing their books, translation tools and recommendation systems can help students reduce grammatical errors and generate ideas and artworks in their digital stories which involve the efforts of language and computer teachers.

Moreover, teachers who are more capable of using AI can empower students' learning using AI-driven technologies (e.g., recommendation systems, intelligent agents)

in their classrooms. They can enable students to apply their multidisciplinary knowledge and help them reach higher cognition levels such as critical thinking, creativity, collaboration, and communication. In higher education, AI literacy education is addressed across different subject areas such as healthcare (Rampton et al., 2020), financial and business (Guerrero-Roldan et al., 2021), and legal education (Demchenko et al., 2021). Higher education educators need to empower their students to use AI applications throughout their life cycle, harness opportunities of using AI technologies in their fields, and be aware of ethical implications and risks (Zawacki-Richter et al., 2019). Concurrently, teachers, regardless of their subject domains, need to develop certain AI digital competencies in order to instruct and assess their students using AI technologies. Ultimately, teachers should be able to guide students on how to manipulate different AI-driven systems and machines that would play a part in their future workplaces.

Nowadays, AI can empower students' reading, writing, and arithmetic abilities (3Rs). For example, e-book systems consist of AI-empowered recommendations that suggest reading materials, practices, and assignments based on students' reading habits (Wu & Peng, 2017). Moreover, AI writing software helps people generate ideas and content for writing articles, novels, blogs, and emails to optimize their work. It can produce compelling articles with the right tone and style, ensure them to be grammatically correct, paraphrase texts, reduce grammatical errors, generate citations automatically, and enhance formatting that improve readability (Nazari et al., 2021; Lin & Chang, 2020). In arithmetic calculation, other applications (e.g., Wolfram Alpha, Symbolab) enable students to take photos of mathematical formulas and questions to generate steps and solve them automatically. However, these applications are too convenient for students to plagiarize in their assignments and examinations (Wu et al., 2021). Teachers should be more sensitive to these AI-enhanced submissions and differentiate whether it is the students' original ideas. It is important for teachers to remind students with correct mindsets and ethical concerns when using these applications to facilitate their learning and living.

Learning and Innovation Skills (4Cs) Scholars have strived the importance of critical, creative thinking, collaboration, and communication in particular to prepare students for today's ever-changing workforce (Van Laar et al., 2017). The skill sets can enable students to create and adapt to changes and encourage students to apply their existing knowledge with peers to solve problems creatively and critically. Studies have shown the effectiveness of using AI technologies to enhance students' learning and innovation skills. For example, when AI is used in music education, it empowers student learning experience through interactive composition, performances, mixing, and appreciation (Zulić, 2019).

Demchenko et al. (2021) identified a digital transformation in legal education that teachers need to equip their students with AI digital competencies. There is a need for law schools to innovate and form stronger interdisciplinary collaboration with AI expertise to enhance their effective professional and everyday activities such as using AI-based tools for criminal justice and identifying algorithmic bias (Deeks, 2019). In business education, AI has a wide range of uses in business to complete

authentic tasks such as aggregating business data, managing customer relationships, and predicting future trends. Business educators need to update their knowledge and enable students to integrate AI into their workplace and create a new user experience for their clients (Williamson & Eynon, 2020; Uzialko, 2022). In medical education, there is a greater demand for digital health applications to promote social distancing during the pandemic. Medical educators need to timely update their knowledge and skills to help their e-patients via AI-driven medical applications such as infection rate prediction, computed tomography image recognition, robotics for patient assessment and drug delivery, and X-ray interpretation (Ahuja & Nair, 2021). These knowledge and skills are useful for teachers and students to become professionals/leaders in their knowledge fields to implement complex cognitive and decision-making tasks and adapt to present scenarios.

Information, Media, and Technology Skills Educators need to prepare themselves to become digital ready so that they are able to teach students related skills such as information, media, and ICT literacy (Gleason & Von Gillern, 2018). Especially in recent years, teachers and students need to adapt to digital transformation and develop related technological skills. In AI-driven classrooms, teachers need to manipulate different AI-enhanced systems to design assessments and examine students' performance using their historical and current data using the adaptive learning system (Guerrero-Roldán et al., 2021). In another study, teachers adopt an automatic mode in an AI-driven service called IBM RXN to enable students to draw target molecules and generate chemical reactions and structure representations (Healy & Blade, 2020). Kostopoulos et al. (2021) mentioned the use of an AI-driven system called DevOps to equip smart city professionals and educators with adequate technological skills to visualize urban innovation in an 11-week online course. Further, teachers need to acquire better technology skills and enable them to intervene early to enhance students' performance and retention (Jokhan et al., 2022). They need to learn how to use different AIED technologies to design their instruction, examine students' learning outcomes, and premise intelligent support for collaborative working. These examples showed that university educators from different disciplines need to equip themselves with technological skills to enable their students to express knowledge, solve problems, and manipulate AI-driven applications so that they are ready to work in AI-driven environments.

Life and Career Skills Life and career skills are important to prepare students to engage as citizens in a dynamic global community and meet different challenges and opportunities in the workforce. Students need to develop positive mindsets, attitudes, and other competencies (e.g., flexibility, adaptability, self-direction, social skills, productivity, responsibility) to navigate complex life and work environments (Van Laar et al., 2017). First, studies suggested that AI has the potential to transform youth employment and students need to develop relevant skills to adapt to this change. For example, Singh et al. (2020) suggested that AI profiling will move away from merely information collection about formal qualifications to a more holistic approach of capturing skills and life experiences. Educators need to upgrade their

students to fit the future job market. Second, improving students' self-efficacy and self-regulation is important when using AI-driven systems to support students' online learning since these systems usually do not consist of a physical teacher to monitor their learning (Guerrero-Roldán et al., 2021). Third, Cetindamar et al. (2022) highlighted four sets of workplace capabilities associated with AI: technological skills (e.g., data collection, analytics, ethics, security), work-related skills (e.g., decision-making, critical thinking, teamwork), human-machine interaction (e.g., situation assessments, affordance analysis, adaptive expertise), and learning-related capabilities (e.g., lifelong learning, self-learning ability). Other studies also mentioned the importance of life and career skills in the fourth industrial revolution such as problem-solving (Mohammed et al., 2021), emotional intelligence, judgment, service orientation, negotiating and cognitive flexibilities (Webber-Youngman, 2017), as well as communication and teamwork skills (Seo et al., 2021). Teachers can enable their students to become adaptive thinkers who equip themselves with technological literacies to solve problems, think critically, lead their teammates, and implement reflective practice (Li & Du, 2017). With these life and workplace skills, students become more digitally ready to contribute to their fields and companies after graduation.

These four essential digital competencies suggest how higher education policymakers consider the necessary educational standards and goals in their universities and schools and provide relevant professional training to develop teachers' readiness in today's AI world. At the classroom level, the framework serves as a guideline to help instructional designers to design suitable curriculum and materials for their students and create positive learning environments. Moreover, such educational standards are important to serve as a basis of educational reform and digital transition to help practitioners to identify the necessary learning outcomes to meet the goals set by the governments, regions, and markets. For example, it helps fulfill job demands in the market, enhance learners' competitiveness, equipping students with futuristic skills, and educate the next generation to become responsible citizens. To achieve these goals, educators need to cultivate their TPACK knowledge to incorporate meaningful learning elements (e.g., assessments, curriculum, instruction) for their students. Since teachers may not be familiarized with these novel technologies, practices for teacher education (e.g., teacher program, professional community, guideline) are important to support educators to equip with necessary knowledge and skills for students' learning and achieve teaching and learning outcomes. Moreover, collaboration from different professionals from higher education institutions, schools, government, industries, and companies could co-design these professional development programs and guidelines to develop meaningful materials, tools, and platforms to support teachers AI literacy education.

10.3 Conclusion

The existing frameworks focused more on students' learning; few of them investigate how to enhance teachers' AI digital competence in professional development programs. However, it is important to promote frontline educators and decision-makers' professional growth so as to improve their pedagogical, content, and technological knowledge to incorporate AI into their classrooms. Research shows that professional development could lead to better instruction, thus improving student learning outcomes (e.g., Kutaka et al., 2017; Gore et al., 2021). Teachers become more capable of connecting their teaching materials to AI and understand the educational standards that guide their teaching, assess, and design students' learning experiences with novel technologies. Further, they can use it wisely to solve teaching problems (e.g., lack of social isolation and motivation in online learning). With AI, teachers can analyze students' behaviors and performance to adapt their teaching and give students immediate assistance to meet learning needs for individual learners.

This Chapter proposed the P21's Framework for 21st Century Learning. Research examples are suggested for each competence to show what teachers should learn before teaching their students. Teachers need to learn technological skills to access AI devices and software (e.g., learning analytic) to improve their teaching and working efficiency (Kexin et al., 2020). On top of technical skills, there is a need to include broader digital competencies such as communication, life and career skills, ethical concerns, teacher identity, attitudes, and mindsets as components of teacher education in AI literacy education (e.g., Akgun & Greenhow, 2021; Seo et al., 2021).

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